INCH-POUND

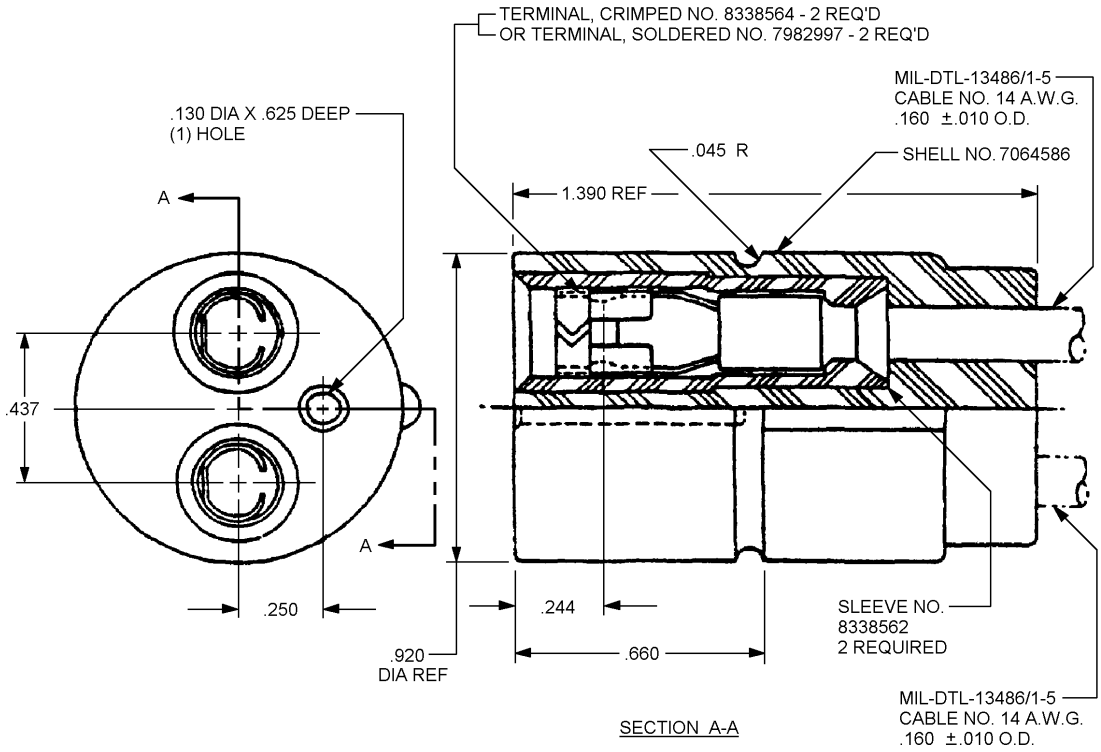
MS27145D
20 March 2018
SUPERSEDING
MS27145C
17 June 2003

DETAIL SPECIFICATION SHEET

CONNECTOR, PLUG, ELECTRICAL, NO. 14 AWG, DOUBLE CONTACT, WATERPROOF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet.



Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.010	0.25	.130	3.30	.244	6.20	.437	11.10	.660	16.76	.92	23.37
.045	1.14	.160	4.00	.250	6.35	.625	15.88	.690	17.53	1.390	35.3

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime - VAI, P.O. Box 3990, Columbus OH 43218-3990 or emailed to CircularConnector@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FIGURE 1. Configuration and dimensions.

AMSC N/A

FSC 5935

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REQUIREMENTS:

Material:

Shell: Rubber in accordance with ASTM-D2000, M3BC610A14C12EO34F19.

Terminal: Brass composition, temper half hard minimum in accordance with ASTM-B36/B36M, ASTM-B121/B121M or ASTM-B124/B124M.

Terminal protective finish: Silver plate, grade D, class S, .0002-inch minimum thickness in accordance with ASTM-B700.

Mating connector: See MS27146.

When connectors are coupled, the following requirements shall apply:

Voltage drop: The voltage drop shall not exceed 20 millivolts when tested in a 28 volt circuit as outlined in MIL-STD-202-307.

Dielectric strength: The assembly shall withstand dielectric strength test of 1000 volts and meet the requirements of MIL-STD-202-302.

Waterproof requirements: The assembly shall be waterproof. Refer to MIL-HDBK-1184, or equivalent, for guidance on waterproof testing.

Salt spray corrosion test: The assembly shall meet the requirements of salt spray corrosion test outlined in MIL-STD-202-101 (5% saline solution, 200 hours duration).

Pull test: Terminal number 8338564 to be crimped to cable. Terminal number 7982997 to be soldered to cable with solder composition SN-50 or SN-60, in accordance with J-STD-004, J-STD-005, and J-STD-006. The terminal assembly shall withstand a pull test of 60 pounds (27 kilograms).

Separation force: The force required to uncouple the connectors shall not exceed 50 percent of the values expressed in the pull test above when tested at -55°C and at 85°C.

Part or Identifying Number (PIN): MS27145-1.

Details of this assembly shall be in accordance with engineering parts list 8589045, which may be obtained from the U.S. Army Tank Automotive Command, Warren, MI 48397-5000.

APPLICATION NOTES:

(These application notes contain information of a general or explanatory nature and are provided for information only.)

Waterproof seal, cable selection and assembly: Recommend that only MIL-DTL-13486/1 cable be used for military applications in order to obtain a waterproof assembly. Other types of cable with smaller outer diameters have been found to be too small to provide a waterproof seal. In addition, to aid in assembling the cable to the connector, assembly lubricants such as isopropyl alcohol or SAE-AS8660, as referenced in Army Pamphlet AMCP 706-360.

Nomenclature: For this specification, connector nomenclature is based on the gender of the connector shell. Single wire connectors with socket contacts specified by this document are commonly referred to as “male connectors”. The associated MS27146 mating connectors with pin contacts are commonly referred to as “female connectors”.

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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. This document references the following:

MIL-DTL-13486/1
MS27146
MIL-STD-202-101
MIL-STD-202-302
MIL-STD-202-307
MIL-HDBK-1184
ASTM-B36/B36M
ASTM-B121/B121M
ASTM-B124/B124M
ASTM-B700
ASTM-D2000
J-STD-004
J-STD-005
J-STD-006
SAE-AS8660
AMCP 706-360
TACOM EPL 8589045

CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – YD
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2018-050)

Review activities:
Army – CR4, MI
Navy – MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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